E-commerce Customer Churn Analysis and Prediction

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1 Motivation

Customer churn is one of the most financially damaging challenges for e-commerce businesses. Because customers can switch platforms easily, it is much more cost-effective to retain existing users than to acquire new ones, with acquisition often costing five to seven times more than retention.

Our goals are to (1) identify which customer interactions and behaviors are most strongly linked to churn, (2) build machine learning models that classify customers as churn or non-churn using historical data, and (3) translate these predictions into actionable insights that support targeted retention strategies and reduce preventable revenue loss.

2 Data Collection and Processing

The dataset is an anonymized e-commerce churn dataset from Kaggle with 5,630 observations and 20 features describing customer behavior, experience, and demographics. The target variable Churn indicates whether a customer has stopped purchasing (1) or remains active (0).

We imputed missing values in key numeric variables (e.g., Tenure, WarehouseToHome, HourSpendOnApp, CouponUsed, OrderCount, DaySinceLastOrder) using median values. Boxplots of tenure, satisfaction score, and days since last order across churn groups helped us visually compare churners vs. non-churners. We saw that churners generally showed shorter tenure, lower satisfaction, and fewer recent interactions (Appendix Figure A1).

Next, we removed the non-predictive identifier variable CustomerID and then applied one-hot encoding to all categorical variables, including PreferredLoginDevice, PreferredPaymentMode, Gender, PreferedOrderCat, and MaritalStatus. Dummy variables were created with *drop_first = True* to prevent multicollinearity in models. After encoding the categorical variables, we split the dataset into training and test sets using an 80/20 stratified split so that the churn ratio stayed consistent in both groups. For models that rely on feature scale—such as logistic regression and LDA—we standardized the numerical features using StandardScaler, fitting on the training data only to avoid data leakage.

These steps produced a clean, consistent dataset suitable for supervised learning while preserving underlying churn patterns.

3 Analytical Models

3.1 Baseline Model

As a benchmark, we used a baseline classifier that always predicts the majority class (non-churn, 0). This model achieved 83.1% accuracy, but never predicted churn (TPR = 0) and had an AUC of 0.5. This means its performance was equivalent to random guessing. Therefore, it serves well as a reference point for evaluating more sophisticated models.

3.2 Logistic Regression

Logistic regression estimates the probability that a customer belongs to the churn (1) or non-churn (0) class. The model outputs probabilities between 0 and 1, which we convert to class labels using a 0.5 threshold. We evaluate performance using AUC to measure ranking ability and a confusion matrix to derive metrics such as accuracy, precision, and recall.

Performance Metrics
Our logistic regression model predicts whether a customer will churn or not with an 88.9% accuracy. This model performs significantly better than the baseline model as it is able to predict positives. In the case of the baseline model, it did not predict any false or true positives, but our model now predicts positives 52% of the time, indicating much higher recall. Additionally, our model's AUC is 0.887, highlighting that our model is a very useful classifier and its ranking ability of the likelihood of churning is effective.

In order to further test sample variation and the stability of our logistic regression model, we performed bootstrap resampling on it. The accuracy we collected ranged from 87 - 91%, meaning that our model remains consistent in performance over resampling. Precision is also strong at 74%, giving us positive predictions we can trust, while recall, on the other hand, averages at 52%, underscoring a moderate sensitivity to positive cases (Appendix Figure B3). In short, our model performs very well but has uncertainty in identifying positives.

3.3 Linear Discriminant Analysis
Linear Discriminant Analysis (LDA) was included as one of our analytical models to evaluate how well a linear decision boundary could separate churners from non-churners. The method works by finding a linear combination of features that maximizes separation between the two groups. This helps LDA consider the combined influence of several features at once, which is valuable when working with datasets that have many variables across different categories. Because LDA depends on covariance structure, we standardized all input features to place them on the same scale. The model was trained on the prepared training data and evaluated on the test set to assess its real-world performance.

Performance Metrics
LDA correctly identifies about 53% of churners, which is significantly better than the baseline model (0% TPR). It maintains a very low false positive rate (3.31%), meaning it rarely misclassifies loyal customers as churn risks. The AUC of 0.884 shows that the model effectively ranks customers by their likelihood of churning, which is valuable for directing retention efforts toward the right individuals.

To assess the stability of our models, we applied bootstrap resampling with 500 samples to estimate variability and confidence intervals. For the LDA model, the bootstrap results showed narrow confidence intervals for accuracy (87.30–91.03%) and FPR (2.25–4.56%), indicating that the model performs consistently across different resampled test sets. Precision also showed relatively tight bounds (68.70–83.15%), suggesting reliable identification of predicted churners. However, the wider interval for TPR (46.14–60.19%) reveals that the model's ability to correctly identify actual churners fluctuates more than other metrics. Overall, the bootstrap analysis suggests that LDA is stable and dependable, but may still miss certain churn patterns in some scenarios (Appendix Figure B3).

3.4 Decision Tree
We used the Decision Tree classifier to evaluate how well a flexible, nonlinear model could capture complex patterns associated with customer churn. Unlike LDA, which assumes linear boundaries, decision trees recursively split the feature space to create highly interpretable rules based on customer attributes. This makes them well-suited for capturing threshold-based behaviors, such as how customers with very low tenure are more likely to churn.

Because unpruned decision trees tend to overfit, we performed cross-validation over a grid of ccp_alpha values (cost-complexity pruning parameters) ranging from 0 to 0.08. For each alpha, a 5-fold cross-validated accuracy score was computed to determine how pruning affected model generalization.

The CV curve showed that accuracy peaked near alpha = 0, and then steadily declined as the tree became more heavily pruned. Based on this analysis, the decision tree with the optimal ccp_alpha = 0 selected by GridSearchCV was used as the final model. (Appendix Figure B1)

Performance Metrics

The results indicate that the tree identifies churners very effectively (TPR > 92%) but also maintains a very low rate of incorrectly flagging non-churners (FPR = 1.60%). The high AUC of 0.953 suggests excellent discrimination ability, meaning the model ranks high-risk customers far better than lower-risk ones. Overall, the decision tree captures nonlinear relationships that simpler models, such as logistic regression or LDA, could not fully detect.

Bootstrap resampling further confirmed the model's stability, producing tight confidence intervals across all key metrics: Accuracy (96.4–98.3%), TPR (87.6–96.2%), FPR (0.9–2.5%), and Precision (87.4–95.7%). These metrics indicate that the decision tree performs consistently across different samples (Appendix Figure B3)**.** Overall, the selected decision tree configuration served as a robust nonlinear model capable of capturing complex patterns in customer churn behavior while maintaining strong and reliable predictive performance.

3.5 Random Forest

We used a Random Forest model to predict customer churn based on randomly selected features from our dataset. It trains a group of decision trees and outputs the mode of class predictions for a classification model. Randomness is incorporated in the differing chosen features for each decision tree, allowing Random Forest models to find complex patterns within data for higher classification accuracy.

In our model, we used GridSearchCV to find the number of features for the max_features hyperparameter that optimizes the model's accuracy. We did this by performing k-fold cross-validation across 5 folds to see what features make the model perform the best. (Appendix Figure B2)

Performance Metrics

The results indicate that our model performed well, achieving an accuracy of 98.2%. It predicts non-churners correctly ~91.6% of the time (TPR) and incorrectly ~0.43% (FPR) of the time. These metrics are comparable to that of the bootstrapped sample: accuracy (98.3%), TPR (91.7%), FPR (0.41%), and precision (97.8%). This shows consistency across model results using the data (Appendix Figure B3). Furthermore, the Random Forest model just about achieves an equivalent accuracy of our decision tree model's upper limit accuracy. This improved performance can be due to the optimization of chosen hyperparameters that allow our model to capture complexities within the most relevant features to churning behavior in our data.

The CV accuracy graph shows that the model showed optimal performance when a maximum of 21 features were chosen for each decision tree of the random forest model (Appendix Figure B2). However, the accuracy was relatively high (>0.945) for a max_features value between 3 and 30, suggesting that a lower number of features can be chosen to decrease the computational complexity of our model without sacrificing high accuracy.

4 Discussion and Synthesis of Results

Across all models, the Random Forest emerged as our best-performing and most practical model for churn prediction. We prioritized precision because it measures the proportion of predicted churners who actually churn, aligning with the goal of avoiding unnecessary spending on customers who are not truly at risk.

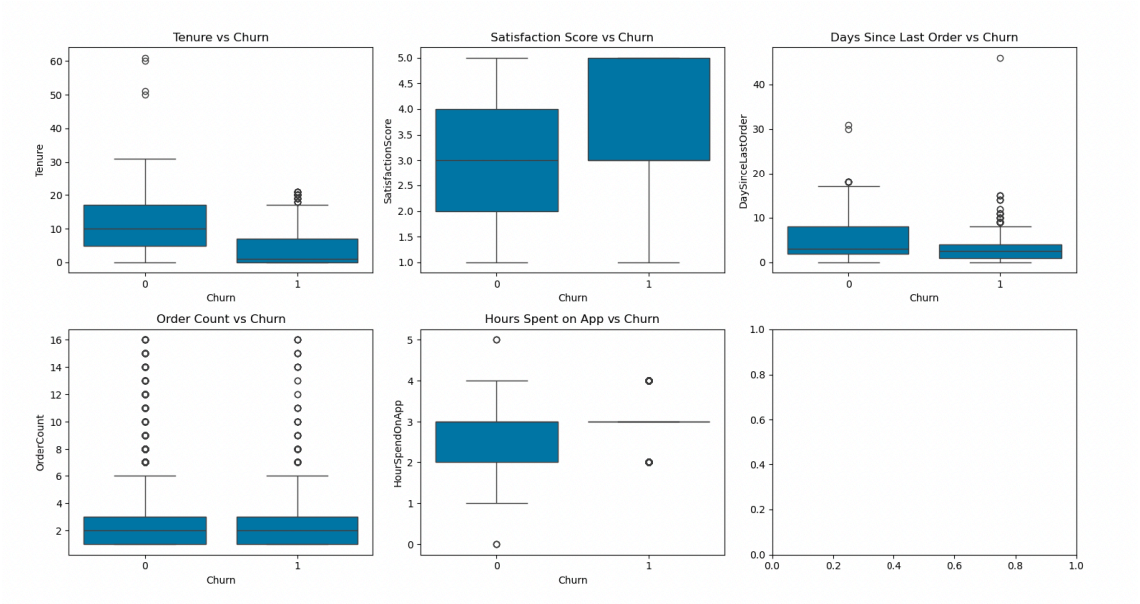
Accuracy	TPR	FPR	PRE	AUC	Model
0.8312611012	0	0	0	0.5	Baseline
0.8889875666	0.5210526316	0.03632478632	0.7443609023	0.8872413405	Logistic Regression
0.973357016	0.9210526316	0.01602564103	0.9210526316	0.9525134953	Decision Tree Classifier
0.9822380107	0.9157894737	0.004273504274	0.9775280899	0.9979644624	Random Forest with CV
0.8925399645	0.5263157895	0.03311965812	0.7633587786	0.8843735942	Linear Discriminant Analysis

Random Forest achieved the highest precision (≈0.98), clearly outperforming the Decision Tree (≈0.92), Logistic Regression (≈0.74), and LDA (≈0.76). At the same time, it also delivered better overall performance across other key metrics—accuracy (98.2%), TPR (91.6%), very low FPR (0.43%), and AUC (0.998). While Logistic Regression and LDA remain more interpretable and still improve substantially over the baseline, they are less precise and sensitive to churners than the tree-based models. The Decision Tree offers strong recall and interpretability but is slightly outperformed by Random Forest on most metrics. Bootstrap analyses further confirm that these results are stable and not driven by a single lucky train-test split, reinforcing Random Forest as the most reliable model for actionable churn prediction in this dataset.

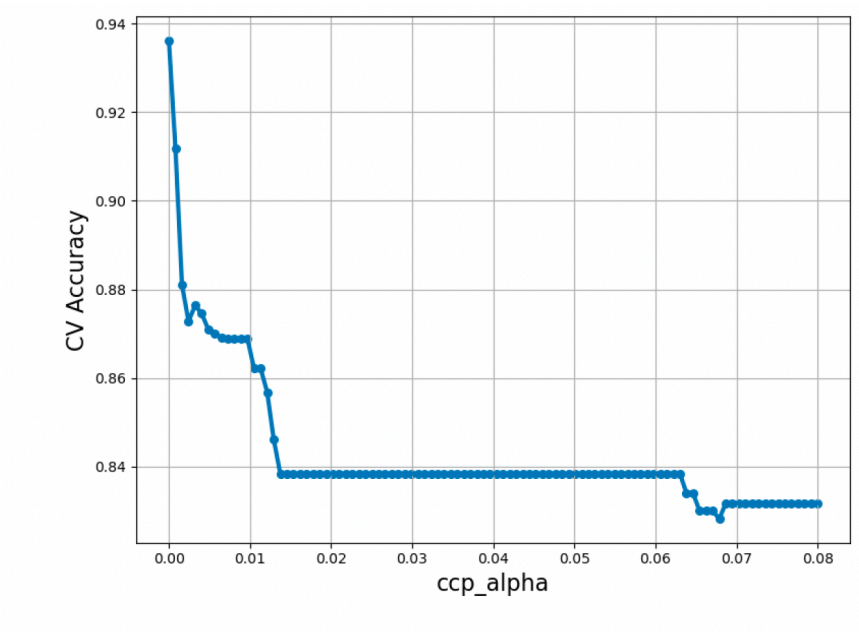
5 Impact of Work

With our best-performing Random Forest model achieving 98.2% accuracy, extremely low false positive rates, and a near-perfect AUC, our churn predictions give e-commerce businesses a solid, data-driven way to act before customers leave. Instead of running broad, generic campaigns, companies can use these churn scores to focus on the customers who are truly at risk and reach them with personalized offers, loyalty perks, or timely service recovery. Because our model links churn risk to concrete drivers like tenure, satisfaction, recency, and coupon usage, managers can also see why customers might leave and focus improvements on things like reliable delivery, customer support, or promotion strategy. At a higher level, these predictions can feed into revenue forecasts, customer lifetime value segmentation, and retention budgeting, helping firms reduce preventable churn and build more stable, long-term revenue.

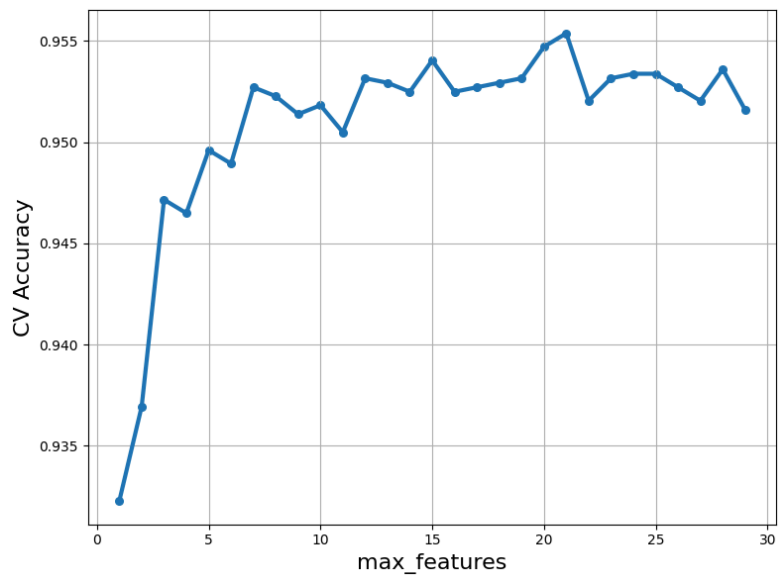
Appendix



Appendix Figure A1 Distribution of Key Variables for Churn vs. Non-Churners



Appendix Figure B1 Cross-Validation Accuracy across ccp_alpha Values for Decision Tree Pruning



Appendix Figure B2 Cross-Validation Accuracy across ccp_alpha Values for Random Forest Pruning

Logistic Model Bootstrap

	metric	mean	std	ci_low	ci_high
0	accuracy	0.889043	0.009546	0.869449	0.908526
1	TPR	0.520998	0.034545	0.452288	0.590593
2	FPR	0.036491	0.006280	0.025079	0.049606
3	precision	0.743022	0.037426	0.675427	0.818185

LDA Model Bootstrap

	metric	mean	std	ci_low	ci_high
0	accuracy	0.892758	0.009515	0.873002	0.910346
1	TPR	0.526740	0.035894	0.461432	0.601878
2	FPR	0.033130	0.005841	0.022505	0.045560
3	precision	0.762923	0.036990	0.687023	0.831518

Tree Model Bootstrap

	metric	mean	std	ci_low	ci_high
0	accuracy	0.973519	0.005056	0.964010	0.983126
1	TPR	0.921077	0.021142	0.875675	0.961767
2	FPR	0.015861	0.004298	0.008542	0.025330
3	precision	0.921543	0.020691	0.873584	0.957453

Random Forest Model Bootstrap

	metric	mean	std	ci_low	ci_high
0	accuracy	0.982554	0.003945	0.974245	0.989343
1	TPR	0.916570	0.019910	0.875634	0.953986
2	FPR	0.004090	0.002105	0.000494	0.008557
3	precision	0.978418	0.011069	0.954179	0.997831

Appendix Figure B3 Bootstrapping results across the models